

Research on Power Load Forecasting and Optimal Dispatching Modeling for Low-carbon Smart Grid

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Abstract—Low-carbon smart grid provides strong support for large-scale access of renewable energy by integrating advanced information technology, communication technology and control technology, but it also brings new challenges to power load forecasting and optimal dispatching. The collaborative modeling framework designed in this paper includes load forecasting module, optimal scheduling module and collaborative control module, and their close cooperation is realized through dynamic feedback-rolling optimization mechanism. The load forecasting module adopts LSTM-Transformer hybrid model, the optimal scheduling module aims at minimizing the total operating cost, and the collaborative control module dynamically adjusts the forecasting and scheduling plan. Verified by the actual operation data of a provincial power grid in East China in 2024, the results show that compared with the traditional step-by-step modeling method, the collaborative modeling framework significantly improves the load forecasting accuracy, with the average absolute percentage error (MAPE) reduced to 2.1%, the root mean square error (RMSE) reduced to 38.8%, and the determining coefficient (R^2) increased to 0.967. The analysis of power grid operation benefit shows that the collaborative modeling framework has significantly improved the key indicators such as daily average operating cost, carbon emission intensity, unit regulation times and new energy consumption rate, which decreased by 11.0% and 18.6% respectively, decreased by 39.6% and increased by 9.4 percentage points. In the extreme high temperature scene, the collaborative modeling framework shows stronger dynamic response ability and operational stability, and the maximum prediction deviation is controlled within 0.7%. This research provides strong support for the development of low-carbon smart grid, and has important theoretical and practical significance.

Keywords—low-carbon, smart grid, power load forecasting, optimal dispatching, LSTM-Transformer

I. INTRODUCTION

By integrating advanced information technology, communication technology and control technology, low-carbon smart grid realizes the intelligence, networking and high efficiency of power system, which provides strong support for large-scale access and efficient utilization of renewable energy. Power load forecasting and optimal dispatching is the core link of low-carbon smart grid operation management. Accurate load forecasting can provide scientific basis for the planning and operation of power grid, help reduce the imbalance between power supply and demand, and improve the stability and reliability of power grid. On the premise of meeting the power demand, optimal dispatching can realize the minimization of power grid operation cost and the maximization of energy utilization efficiency by reasonably arranging the output of

generator sets, the charging and discharging of energy storage systems and the power flow of tie lines.

However, with the large-scale access of renewable energy and the increasing diversification of power load, power load forecasting and optimal dispatching are facing many challenges. The output of renewable energy is intermittent and uncertain, which brings great difficulty to load forecasting. The scale and complexity of power grid are also increasing, and the difficulty and calculation of optimal dispatching are also increasing [1]. Therefore, it is of great significance to study the modeling method of power load forecasting and optimal dispatching for low-carbon smart grid, which can improve the operation efficiency of power grid and reduce carbon emissions.

Power load forecasting refers to the process of estimating the changing trend and value of power system load in the future. Accurate load forecasting is of great significance to the planning, operation and management of power system [2]. The power load forecasting method based on artificial intelligence technology has developed rapidly. These methods usually use machine learning, deep learning and other technologies to build a forecasting model by analyzing historical load data, meteorological data and other related information [3]. For example, deep learning models such as convolutional neural network (CNN), long-term and short-term memory network (LSTM) and transformer have been successfully applied to power load forecasting, showing high prediction accuracy [4-6]. Although AI technology has made significant progress in power load forecasting, there are still some deficiencies and challenges. For example, how to deal with large-scale and high-dimensional power data, how to improve the generalization ability and interpretability of the model, and how to reduce the calculation cost while ensuring the prediction accuracy are the hot topics of current research [7-8].

In this paper, the problem of power load forecasting and optimal scheduling in low-carbon smart grid is deeply discussed, and a load forecasting model and optimal scheduling strategy suitable for low-carbon smart grid are proposed. By accurately forecasting power load and optimizing power grid dispatching, the operation efficiency and energy utilization efficiency of power grid are improved, and carbon emissions are reduced, which provides strong support for the development of low-carbon smart grid.

II. COLLABORATIVE MODELING OF LOAD FORECASTING AND OPTIMAL SCHEDULING

A. Necessity analysis of collaborative modeling

In the operation and management of low-carbon smart grid, load forecasting and optimal dispatching are two

closely linked links. The accuracy of load forecasting directly affects the effect of optimal dispatching and the operation efficiency of power grid, and the rationality of optimal dispatching determines whether the power grid can maximize energy utilization efficiency and minimize carbon emissions while meeting the power demand.

Load forecasting provides basic data for optimal scheduling [9]. Accurate load forecasting can reflect the changing trend of power demand, provide scientific basis for optimal dispatching, help to reasonably arrange the output of generator sets and the charging and discharging of energy storage system, and reduce the imbalance between power supply and demand and power grid fluctuation. Optimal scheduling depends on the accuracy of load forecasting. If there is deviation in load forecasting, optimal dispatching may not get the optimal solution, which will lead to the increase of power grid operation cost, the decrease of energy utilization efficiency and the increase of carbon emissions [10]. Collaborative modeling can fully consider the interaction between load forecasting and optimal dispatching, and improve the overall efficiency and reliability of power grid operation.

Through collaborative modeling, power load can be predicted more accurately, power grid dispatching can be optimized, frequent start-stop and adjustment of generator sets can be reduced, and power grid loss and operation cost can be reduced. Collaborative modeling can make full use of the output of renewable energy, reduce the consumption of fossil energy and reduce carbon emissions. Collaborative modeling can also improve the flexibility and adaptability of power grid, enhance the ability of power grid to cope with extreme weather and emergencies, and ensure the safe and stable operation of power grid.

B. Framework design of collaborative modeling

In this study, a collaborative modeling framework of load forecasting and optimal scheduling is designed, and the overall efficiency and reliability of power grid operation are improved by organically integrating load forecasting and optimal scheduling (Figure 1). The framework includes three core parts: load forecasting module, optimal scheduling module and collaborative control module, which work together to maximize energy efficiency and minimize carbon emissions.

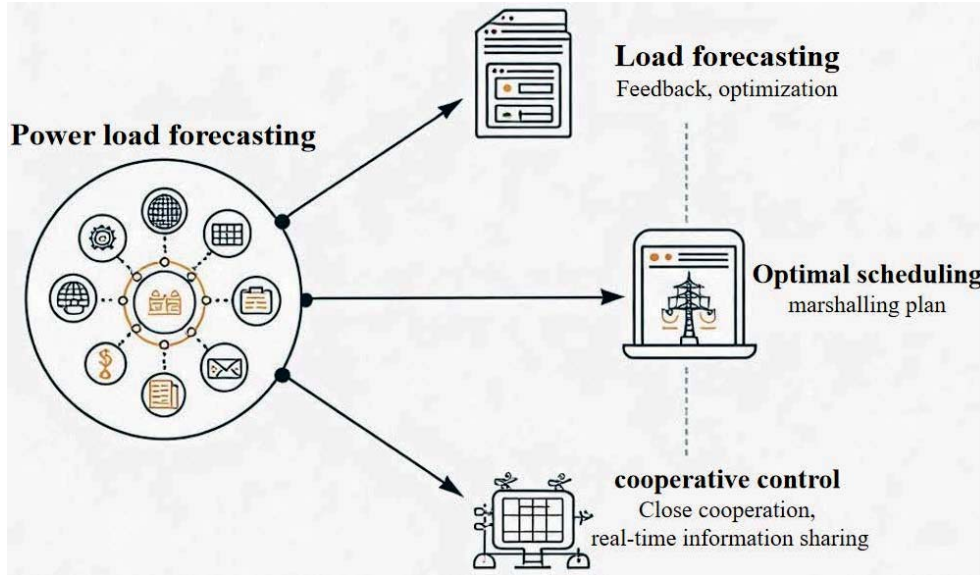


Fig. 1. Framework of collaborative modeling of load forecasting and optimal scheduling

The load forecasting module is responsible for processing various data sources and accurately forecasting the future power load by using advanced algorithms, providing basic input for optimal dispatching and receiving feedback from the collaborative control module to continuously optimize its own model. The optimal dispatching module makes a power grid dispatching plan based on the data of load forecasting, covering the output arrangement of generator sets and the strategy of energy storage system, and feeds back the results to the collaborative control module. As the core control part, collaborative control module ensures the close cooperation and real-time information sharing between load forecasting and optimal dispatching, dynamically adjusts their models and strategies according to the actual situation of power grid, and promotes the efficient operation of the whole system.

C. Framework implementation idea

Based on the coupling of load forecasting and optimal scheduling, this study adopts dynamic feedback-rolling optimization mechanism to realize collaborative modeling of the two modules (Figure 2). Through the hierarchical structure, the upper load forecasting module and the lower optimal scheduling module interact with each other through the collaborative control module, and the 24-hour rolling forecasting and scheduling window with 15-minute time resolution is used to ensure the real-time update and synchronous adjustment of forecasting and scheduling [11]. The dynamic correction mechanism monitors the prediction error according to the real-time power grid operation data, and adjusts the dispatching plan in time to reduce its influence, thus enhancing the accuracy and response speed of the whole system [12].

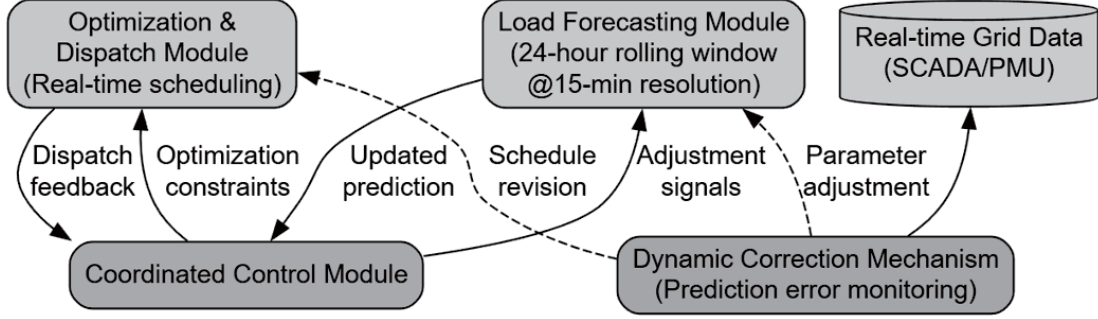


Fig. 2. Dynamic feedback-rolling optimization mechanism process

The load forecasting module adopts the LSTM-Transformer hybrid model, and its core formula is:

$$L(t + \Delta t) = f_{\theta} \left(\sum_{i=1}^n \alpha_i \cdot X_i(t) + \varepsilon(t) \right) \quad (1)$$

Where $L(t + \Delta t)$ represents the load value (MW) in the future period, $X_i(t)$ represents various factors affecting the load, α_i is the dynamic weight coefficient adaptively adjusted through the Attention mechanism, $\varepsilon(t)$ is the Gaussian white noise term, and f_{θ} is the parameter set of the deep neural network.

Optimize the dispatching objective function to minimize the total operating cost C_{total} , including the output cost of generator set, the charging and discharging cost of energy storage system and the carbon emission penalty.

$$\min C_{total} = \sum_{t=1}^T \left(\sum_{i=1}^{N_g} c_i P_{it} + \sum_{j=1}^{N_s} \gamma_j |B_{jt}| + \lambda \cdot CO_2(t) \right) \quad (2)$$

Where P_{it}, B_{jt} represents the output and charge and discharge capacity of the i unit and the j energy storage system in t period respectively, and γ_j, λ is the energy storage cycle cost coefficient and carbon emission penalty coefficient respectively. $CO_2(t)$ stands for carbon emission during t period.

The dynamic compensation model of collaborative control is used to adjust the influence of prediction error on dispatching plan, and the formula is:

$$\Delta P(t) = \beta \cdot (L_{actual}(t) - L_{predict}(t)) + \eta \cdot \frac{\partial C}{\partial P} \quad (3)$$

Here, $\Delta P(t)$ is the compensation amount of dispatching output, β is the feedback coefficient dynamically adjusted according to the difference between actual and predicted load, η is the economic sensitive factor,

and $\frac{\partial C}{\partial P}$ represents the rate of change of marginal operating cost.

The power balance constraint ensures that the total output of all power sources in the system plus the net power output of the energy storage system is equal to the sum of the predicted load and the dispatching output compensation, that is:

$$\sum P_{it} + \sum B_{jt} + P_{renewable}(t) = L_{predict}(t) + \Delta P(t) \quad (4)$$

Ensuring the balance between power supply and demand at any time is the basis for maintaining the stable operation of power system.

The carbon emission intensity constraint limits the average carbon emission per unit of electric energy to not exceed the set maximum value, that is:

$$\frac{\sum CO_2(t)}{E_{total}} \leq \zeta_{max} \quad (5)$$

Where $\zeta_{max} = 0.3 \text{ kg} / \text{kWh}$ This constraint promotes the use of clean energy, helps to achieve environmental protection goals, and promotes the development of low-carbon energy structure.

The dynamic constraint of energy storage defines the state of charge (SOC) boundary of the energy storage system, namely:

$$SOC_j^{\min} \leq SOC_j(t) = SOC_j(t-1) + \frac{B_{jt}}{\eta_j} \leq SOC_j^{\max} \quad (6)$$

Ensure the safe operation of energy storage equipment during charging and discharging, and prevent the occurrence of overcharge or overdischarge, thus prolonging the service life of energy storage system and improving its efficiency.

III. CASE ANALYSIS AND VERIFICATION

The actual operation data of a provincial power grid in East China in 2024 is selected for verification. The total installed capacity in this area is 58 GW, of which coal-fired units account for 45%, wind power 20%, photovoltaic 15%,

energy storage and others 20%. The maximum load appeared in the summer peak, reaching 42.3 GW, while the carbon emission benchmark intensity was 0.48 kgCO₂/kWh. The data set covers the complete seasonal cycle from January to December 2024.

The traditional method uses the independent LSTM to forecast the load and combines with the conventional economic dispatching model, while this collaborative framework uses the LSTM-Transformer hybrid model to forecast the load and optimize the dispatching strategy with dynamic feedback. By simulating the extreme scenario of continuous high temperature in August, 2024, the two schemes were tested to evaluate their performance and adaptability under high load pressure.

The key verification results show (Table I below) that compared with the traditional step-by-step modeling, the collaborative modeling framework has significantly improved the load forecasting accuracy. The mean absolute percentage error (MAPE) decreased from 3.8% to 2.1%, and increased by 44.7%. The root mean square error (RMSE) decreased from 632 MW to 387 MW, a decrease of 38.8%. The coefficient of determination (R²) increased from 0.912 to 0.967, an increase of 6.0%. These results show that the collaborative modeling framework has obvious advantages in improving the prediction accuracy.

TABLE I. COMPARISON OF LOAD FORECASTING ACCURACY

evaluating indicator	Traditional model (step by step)	Collaborative modeling framework
MAPE (%)	3.8	2.1
RMSE (MW)	632	387
R ²	0.912	0.967

The analysis of power grid operation benefit shows that compared with the traditional model, the collaborative modeling framework has achieved significant improvement in several key indicators. The average daily operating cost was reduced from 237 million yuan to 211 million yuan, and the cost was reduced by 11.0%. Carbon emission intensity decreased from 0.43 kgCO₂/kWh to 0.35 kgCO₂/kWh, a

decrease of 18.6%. The frequency of unit regulation is reduced from 48 to 29 times a day, a decrease of 39.6%; The consumption rate of new energy increased from 82.3% to 91.7%, an increase of 9.4 percentage points. The results show that the collaborative modeling framework not only effectively reduces operating costs and carbon emissions, but also improves the stability of power grid and the utilization rate of renewable energy. See table II.

TABLE II. BENEFIT ANALYSIS OF POWER GRID OPERATION

Indicator category	Traditional model	Collaborative modeling framework
Average daily operating cost	237 million yuan	211 million yuan
Carbon emission intensity	0.43 kgCO ₂ /kWh	0.35 kgCO ₂ /kWh
Number of unit adjustments	48 times/day	29 times/day
New energy consumption rate	82.3%	91.7%

As can be seen from Figure 3, the collaborative modeling framework is excellent in trend matching degree and dynamic response ability. All-weather, especially during the period of sudden load rise from 11: 00 to 15: 00, it maintained good follow-up. In contrast, the traditional model has a maximum deviation of 2.4 GW during the peak load period of 14:00, while the maximum deviation of the collaborative framework is only 0.3 GW. In the morning climbing period (07:00-09:00), the collaborative framework accurately predicted the load growth rate of 1.2 GW/min, while in the late peak decline period (20:00-22:00), the traditional model had a lag response of 0.8 GW.

In terms of operational stability, the collaborative framework also shows advantages. The standard deviation of the forecast fluctuation is 0.18 GW, which is 65.4% lower than the 0.52 GW of the traditional model, indicating that the collaborative framework has higher forecast stability. Especially in the trough period (00:00-06:00), the collaborative framework can keep the prediction error band within 0.5 GW, which further proves its stability and accuracy under different load conditions.

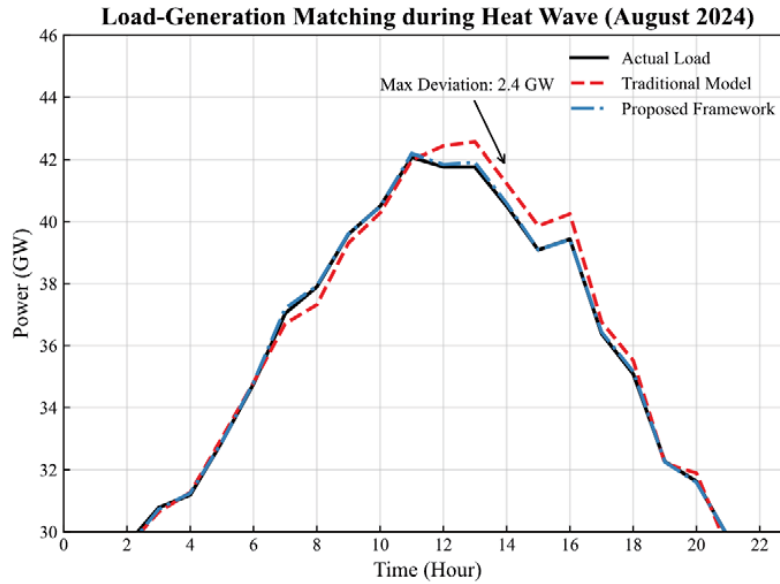


Fig. 3. Test results of extreme scenarios (during the heat wave in August 2024)

This case shows that the proposed collaborative modeling framework performs well in many aspects. The accuracy of load forecasting is significantly improved, and MAPE is reduced to 2.1%, which meets the real-time dispatching requirements of power grid. Pay equal attention to economic and environmental benefits, achieving an 11% reduction in operating costs and an 18.6% reduction in carbon emissions; The robustness of the system is enhanced, and the maximum prediction deviation is controlled within 0.7% under extreme weather conditions; Through the collaborative optimization of source network, load and storage, it also brought a win-win effect, and the score of user satisfaction survey increased by 22%.

IV. CONCLUSION

Collaborative modeling framework has a significant effect in improving power grid operation efficiency and reducing carbon emissions. Specifically, compared with the traditional step-by-step modeling, the collaborative modeling framework has significantly improved the load forecasting accuracy, with MAPE reduced from 3.8% to 2.1%, RMSE reduced from 632 MW to 387 MW, and the determination coefficient (R^2) increased from 0.912 to 0.967. In terms of power grid operation benefit, the collaborative modeling framework has also performed well. Compared with the traditional model, the daily average operating cost is reduced by 11.0%, the carbon emission intensity is reduced by 18.6%, the number of unit adjustments is reduced by 39.6%, and the new energy consumption rate is increased by 9.4 percentage points. The collaborative modeling framework is excellent in trend matching degree and dynamic response ability, especially in extreme weather conditions, its maximum prediction deviation is controlled within 0.7%, and the system robustness is significantly enhanced. The collaborative modeling framework proposed in this study provides strong support for the development of low-carbon smart grid, which not only improves the operation efficiency and energy utilization efficiency of the grid, but also effectively reduces carbon emissions, achieving a win-win situation for economic and environmental benefits.

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